SUMANTH MALIPEDDI

Email: sumanth.malipeddi97@gmail.com
Portfolio: sumanthmalipeddi.com
India (Remote / Open to Relocate)
linkedin.com/in/sumanth-malipeddi | github.com/sumanthmalipeddi

SUMMARY

Data Engineer and AI systems builder who owns production data infrastructure end-to-end at a B2B SaaS startup — ETL/ELT pipelines (Airflow, dbt, PostgreSQL, MongoDB), production-scale loads with zero-record-loss guarantees (811K+ records), entity resolution across heterogeneous sources, and multi-agent LLM research systems. MS in Data Science & AI (9.5/10 GPA, IISER Tirupati). Ships complete products solo: a native iOS media player live on the App Store in 175 countries, a fintech reconciliation app, and an LLM health platform. Pivoted from civil engineering, where a self-built GCP housing-analytics pipeline over 4 years of government data catalyzed the transition. Targeting roles at the intersection of data engineering, machine learning, and AI systems.

WORK EXPERIENCE

- **GetMySaas (B2B SaaS Startup)** — Remote
  Associate Data Science Engineer — Sole Data Infrastructure Owner — *Oct 2025 – Present*
  - **Own the company's canonical market taxonomy**: 166 categories, architect-approved v3.6.0 — classified 423,163 Product Hunt products against it with zero record loss; designed the governance layer: schema validators, machine-parseable routing overrides, and watch registers with falsifiable review triggers.
  - **Engineered production-scale loads with zero-record-loss guarantees**: loaded 811,548 person records from production AWS MongoDB over a load balancer that kills long-lived cursors, using chunked `_id` pagination with retry; staging tables + count-reconciliation gates + atomic content swap (content-swap over rename because Postgres views bind by OID).
  - **Built golden-record and outreach marts**: `dim_companies` (48,729) and `dim_vendor_contacts_outreach` (792,816) with person-level dedup and honest-schema design — no synthetic identity keys, `is_deliverable` gated FALSE until independently verified; 10/10 dbt tests passing.
  - **Implemented the dbt transformation framework**: 12+ models across staging/intermediate/marts, 32 schema and custom data tests, SCD Type 2 snapshots, seed-driven configuration (domain overrides, country normalization), and 5 reusable macros — orchestrated via a 9-task Airflow DAG with Cosmos in a 6-service Docker Compose deployment.
  - **Designed multi-source entity resolution**: multivariate match-candidate scoring across 6 heterogeneous sources (5-dimension confidence model), 14-column source-attribution provenance, dynamic regex revenue parsing, and automated same-name-different-domain quality detection.
  - **Built multi-agent LLM research systems as production tooling**: workflows up to 54 parallel agents (647 web lookups) with adversarial researcher-vs-skeptic verification, structured outputs, resume-from-cache, and model-tiering for cost control; a quantitative inclusion framework (3 hard gates, 10 weighted dimensions) turned research into architect-approved decisions.
  - **Advanced SQL at scale**: JSONB unpacking over 423K documents (`jsonb_array_elements`, nested extraction), `DISTINCT ON`, window functions, CTEs, `percentile_cont`, `array_agg` with `FILTER`; `psycopg2 execute_values` bulk loading.
- **Engineering Secretary — Project & Data Analytics** — Tirupati, India
  Project Engineer — *Oct 2019 – Jun 2024*
  - **Led quantitative project analytics**: managed construction data across 200+ residential units — material consumption tracking, cost variance analysis, schedule adherence metrics, and workforce productivity models; reduced cost overruns by 15%.
  - **Built Housing Data Analytics Platform (career-pivot project)**: end-to-end GCP pipeline using Python, Mage (ETL orchestration), BigQuery, and Power BI — curated 4 years of housing data from revenue, municipal, and administration departments into a dimensional model with stakeholder dashboards.
  - **Cross-functional stakeholder management**: coordinated architects, contractors, government bodies, and clients; managed teams of 20–50 across concurrent projects — translating technical specifications into business decisions.

KEY PROJECTS

- **MaxMediaPlayer — Native iOS Video Player (Live on the App Store)** — Swift 6, VideoToolbox, StoreKit 2
  Solo: custom playback engine, App Store release across 175 countries — *Jul 2026*
  - **Systems-level media engineering**: wrote a pure-Swift Matroska/WebM (EBML) demuxer from scratch feeding a bounded-memory streaming engine — memory-mapped reads into Apple's hardware decoders (AV1/HEVC/H.264 via VideoToolbox) playing 60GB files with instant start; O(1) seeking via Cues index.
  - **Release and ops automation**: automated the entire App Store operation via the App Store Connect API — subscription/IAP creation across 175 territories, localized listings in 5 languages, and review submission as a CI-style pipeline; passed App Review including LGPL-2.1 compliance.
  - **Engineering rigor**: 327 unit tests on the pure-logic core running on macOS + Linux CI under Swift 6 strict concurrency, in a Tuist-managed monorepo (∼20 SwiftPM packages); fixed field-reported beta issues across 9 builds in 3 weeks.
- **PaySure — Payment Reconciliation for Gig Workers** — Next.js 16, TypeScript, SQLite/Drizzle
  Solo: full-stack fintech app from self-written product spec to ship — *2026*
  - **Full-stack build**: 12 routes, 59 unit tests, OAuth 2.0, PDF/CSV export — converts Google Calendar events into employer-format payout statements and reconciles them against actual payments, flagging underpaid hours and TDS deductions.
  - **Fintech-grade correctness**: timezone-correct pay-period engine for India's 26th–25th cycles (UTC/IST boundary tests), integer-paise money arithmetic, three-bucket reconciliation, and TDS-aware (§194J) verification with Form 26AS ledger.
  - **Security-first integration**: OAuth read-only scope with AES-256-GCM-encrypted refresh tokens and CSRF state validation; calendar descriptions deliberately never stored; from-scratch ICS parser with RRULE expansion as no-OAuth fallback.

- **AharaOS — Condition-Aware Health OS** FastAPI, Supabase, Gemini + Claude, SwiftUI
  Solo: LLM product with multi-provider failover, deployed on Fly.io *2026*
  - **AI health platform**: 120+ REST endpoints, 40 SQL migrations, RLS on all 19 tables — generates condition-aware nutrition protocols with medical-condition filters as hard gates before macro ranking; multimodal LLM extraction of 92 markers from lab-report PDFs.
  - **Multi-provider LLM engineering**: Pydantic-schema structured outputs with automatic Gemini→Claude failover (validated during a real provider outage), per-user cost caps with a DB-driven pricing ledger, and defense-in-depth security (SSRF guard, CSP+nonce, rate limiting) behind 6 CI gates.

TECHNICAL SKILLS

- **Data Engineering**: Apache Airflow (Cosmos/Astronomer, custom operators), dbt (tests, snapshots, seeds, macros), Airbyte, ETL/ELT design, entity resolution, medallion architecture, SCD Type 2, zero-record-loss load patterns, Kafka (fundamentals; hands-on in progress)
- **Programming & SQL**: Python (Pandas, psycopg2, FastAPI, Pydantic), SQL (PostgreSQL, BigQuery — window functions, CTEs, JSONB, `DISTINCT ON`, query optimization), MongoDB (pymongo), Swift 6, TypeScript
- **AI / ML**: Multi-agent LLM orchestration (adversarial verification, structured outputs, model tiering, cost engineering), Gemini & Claude APIs, TF-IDF similarity, quantitative evaluation frameworks; ML/DL foundations (Scikit-learn, TensorFlow, statistical modeling)
- **Cloud & Infrastructure**: AWS (S3, EC2, IAM), GCP (Cloud Composer, BigQuery, GCS), Snowflake, Docker & Docker Compose, GitHub Actions CI/CD, Fly.io, Linux, Git/GitHub PR workflow
- **Data Quality & Governance**: dbt tests, honest-schema design, count-reconciliation gates, validation frameworks, taxonomy governance, pipeline monitoring

EDUCATION

- **Indian Institute of Science Education and Research (IISER)** Tirupati, India
  Master of Science – Data Science & Artificial Intelligence; GPA: 9.5/10.0 *Jul 2024 – Jul 2025*
  - **Coursework**: Machine Learning, Deep Learning, NLP, Statistical Modeling, Big Data Analytics, Database Systems, Cloud Computing, Data Structures & Algorithms
- **SASTRA University** Thanjavur, India
  Bachelor of Technology; GPA: 8.54/10.0 *Jul 2014 – May 2018*

CERTIFICATIONS

- **dbt Fundamentals**: dbt Labs — Data modeling, testing, documentation, materializations
- **Apache Airflow 3 Fundamentals + DAG Authoring**: Astronomer — Orchestration, TaskFlow API, advanced patterns
- **Snowflake Hands-On Essentials**: Snowflake — Data warehousing, SQL analytics
- **AWS Cloud Practitioner Essentials**: AWS — Cloud architecture, core services

LEADERSHIP & COMMUNITY

- **Teaching Assistant — Data Structures, Python & AI/ML** Masai School
  Technical Mentor — teach DSA, Python, and ML to aspiring engineers *2025 – Present*
- **Data Science Learning Community — @the.intlovert** 4300+ learners
  Built educational platform, 50+ mentoring sessions via Topmate; 1,100+ day daily learning streak *Aug 2025 – Present*